

ML Statistics Theory Notes

Descriptive Statistics

Mean, median, mode, variance, standard deviation, skewness, kurtosis.

Probability

Conditional probability, Bayes theorem, joint and marginal distributions.

Distributions

Normal, Bernoulli, Binomial, Poisson, Exponential.

Hypothesis Testing

Null/alternative hypothesis, p-value, t-test, chi-square, ANOVA.

Correlation & Covariance

Pearson correlation measures linear relation; covariance direction.

Sampling

Random, stratified, bootstrap, cross-validation.

Regression

Linear regression assumptions, bias-variance, residuals.

Feature Statistics

Normalization, standardization, outlier detection using IQR/Z-score.